



Standard Treatment Workflow (STW)

ANTENATAL CORTICOSTEROIDS FOR PRETERM BIRTH

INTRODUCTION

Antenatal corticosteroid (ACS) therapy is one of the most cost-effective interventions for improving outcomes of preterm birth. When given to women with a high likelihood of preterm birth, it reduces neonatal mortality, incidence and severity of respiratory distress syndrome, and intraventricular haemorrhage

WHEN TO GIVE

Pregnant women at 24⁺⁰ to 33⁺⁶ weeks of gestation with a high likelihood of preterm birth within the next 7 days

ELIGIBILITY CRITERIA

GIVE ACS WHEN ALL CONDITIONS ARE MET IN WOMEN AT 24⁺⁰ TO 33⁺⁶ WEEKS OF GESTATION:

- High likelihood of preterm birth within the next 7 days due to any one:
 - Spontaneous preterm labour: regular contractions with cervical change consider (dilatation or effacement)
 - PPROM without clinical infection
 - Antepartum Hemorrhage
 - Severe pre-eclampsia/eclampsia
 - Planned preterm birth for maternal/fetal indication
- Gestational age assessed accurately, preferably by early USG
- No clinical chorioamnionitis or systemic maternal infection
- Adequate childbirth care is available at the facility or at the referral facility
- Adequate preterm newborn care be ensured(including CPAP) at the facility or the receiving facility

IMPORTANT: Start ACS promptly once eligibility criteria are met, even if the full course may not be completed before birth

DRUG & DOSE



DEXAMETHASONE SODIUM PHOSPHATE 6mg IM EVERY 12 HOURS X 4 DOSES

- Dexamethasone is preferred: it is effective, low-cost, widely available and more heat-stable
- Betamethasone regimen (12 mg every 24 hours x 2 doses) requires an appropriate acetate + phosphate preparation, which is not available in India. Betamethasone phosphate alone is shorter acting than dexamethasone

WHEN NOT TO GIVE

- Clinical chorioamnionitis or systemic maternal infection
- Preterm birth within next 7 days is unlikely
- Routine use at ≥34 weeks
- More than one repeat course

SPECIAL SITUATIONS

- When otherwise eligible, do not withhold ACS because of: multiple pregnancy, fetal growth restriction, hypertensive disorders, or diabetes in pregnancy
- In diabetes: monitor and optimize maternal glucose control

WHEN TO GIVE REPEAT COURSE

- Give ONE repeat course only if ALL criteria are met; neuro-developmental
 - GA 24⁺⁰ to 33⁺⁶ weeks
 - High likelihood of preterm birth within the next 7 days
 - The previous ACS course was started ≥7 days earlier
- Do NOT give more than one repeat course (More than one repeat course may cause fetal harm, and is not recommended)

DOCUMENTATION

- Document: gestational age, indication, drug/dose, date and time, initial/repeat course, next dose due
- Record in: case sheet, MCP card, and referral slip

REFERRAL/TRANSFER

- Pregnant women with likelihood of early preterm birth should be referred to a facility (in-utero transfer) with at least level II care (CPAP) for preterm infants
- Level I facilities can give the first dose of ACS before referral, provided the pregnant woman meets the eligibility criteria, but referral/transfer should not be delayed to complete the course

DOs

- Confirm accurate GA and that preterm birth is likely within next 7 days
- Start ACS promptly once eligibility criteria are met
- Monitor maternal glucose closely in women with diabetes
- Document dose timing and arrange in-utero transfer when higher-level care is required

DONTs

- Do NOT give ACS when preterm birth is unlikely within next 7 days
- Do NOT delay an indicated birth or referral to complete the ACS course
- Do NOT give ACS in clinical chorioamnionitis or systemic maternal infection
- Do NOT routinely use ACS at ≥34⁺⁰ weeks or give more than one repeat course

KEY PERFORMANCE INDICATORS (KPIs)

ACS coverage = $\frac{\text{Women at 24+0–33+6 weeks identified with high likelihood of preterm birth who received } \geq 1 \text{ ACS dose}}{\text{All women at 24+0-33+6 weeks identified with high likelihood of preterm birth}} \times 100$ (Target: ≥80%)

Appropriate ACS use = _____ (Target: ≥95%)

TIMELY ACS IMPROVES PRETERM OUTCOMES — GIVE WHEN ELIGIBILITY CRITERIA ARE MET AND BIRTH IS LIKELY WITHIN 7 DAYS

RELATED INFORMATION



Video on Antenatal Steroids



Antenatal Steroid Use

ABBREVIATIONS

ACS: Antenatal Corticosteroids
GA: Gestational Age
PPROM: Preterm Prelabour Rupture of Membranes
CPAP: Continuous Positive Airway Pressure
MCP: Mother and Child Protection Card

IM: Intramuscular
IVH: Intraventricular Hemorrhage
RDS: Respiratory Distress Syndrome
USG: Ultrasonography

REFERENCES

McGoldrick E, et al. Antenatal corticosteroids for accelerating fetal lung maturation for women at risk of preterm birth. Cochrane Database Syst Rev. 2020;12:CD004454.
World Health Organization. WHO recommendations on antenatal corticosteroids for improving preterm birth outcomes. Geneva: WHO; 2022.

This STW has been prepared by national experts of India with feasibility considerations for various levels of healthcare system in the country. These broad guidelines are advisory, and are based on expert opinions and available scientific evidence. There may be variations in the management of an individual patient based on his/her specific condition, as decided by the treating physician. There will be no indemnity for direct or indirect consequences. Kindly visit the DHR portal for more information (stw.icmr.org.in) © Department of Health Research, Ministry of Health & Family Welfare, Government of India.